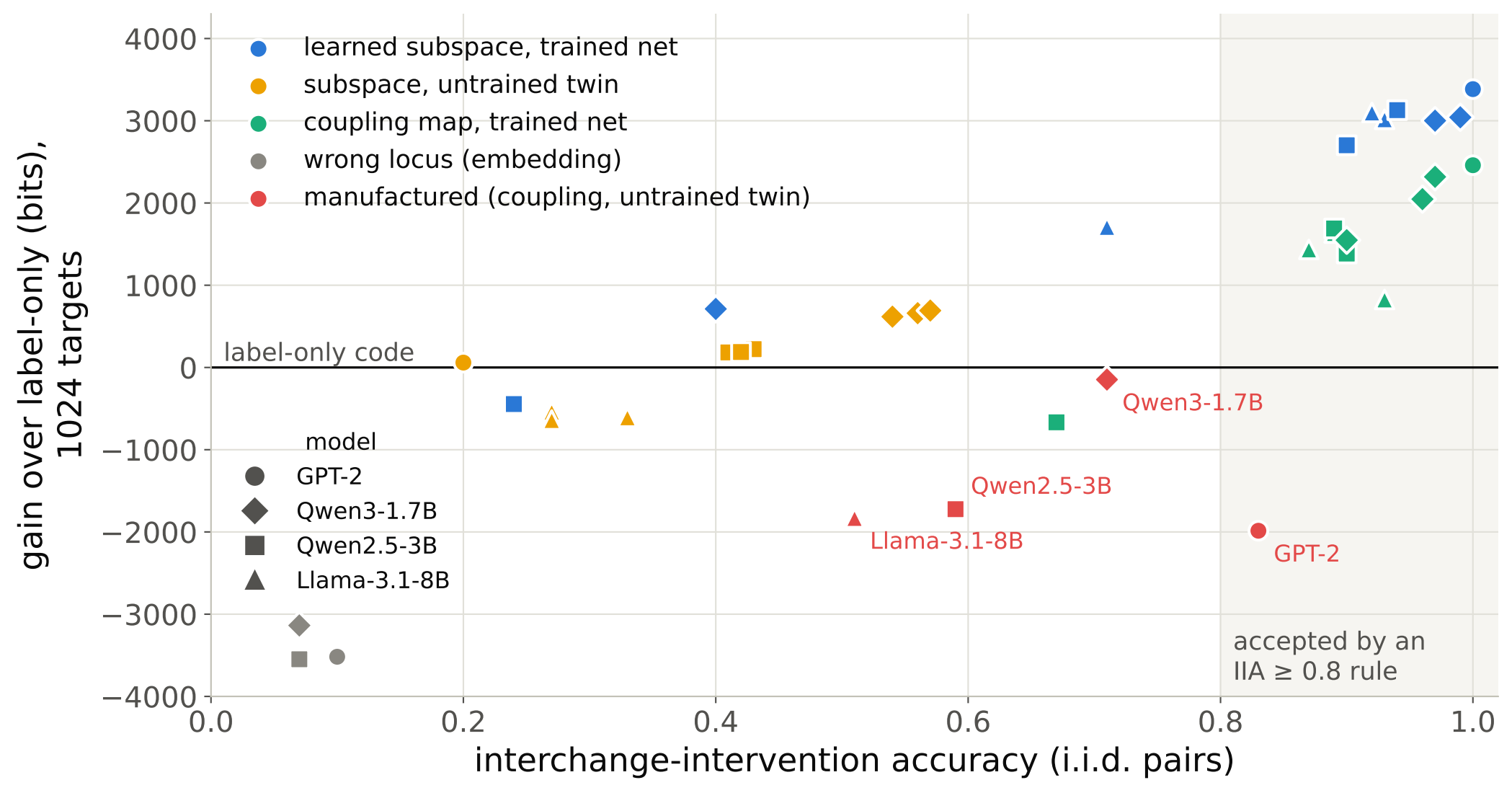
learned subspace, trained net
subspace, untrained twin
coupling map, trained net
wrong locus (embedding)
manufactured (coupling, untrained twin)
label-only code
model
GPT-2
Qwen3-1.7B
Qwen2.5-3B
Llama-3.1-8B
Qwen3-1.7B
Qwen2.5-3B
Llama-3.1-8B
GPT-2
accepted by an
IIA ≥ 0.8 rule
gain over label-only (bits),
1024 targets
interchange-intervention accuracy (i.i.d. pairs)
4000
3000
2000
1000
0
−1000
−2000
−3000
−4000
0.0
0.2
0.4
0.6
0.8
1.0